Check: vabs(-4) returns 4.

4. Back · Flash [NUM] Write in decimal notation 2.5×10^3 then 6.2×10^{-2}.

Ans. 2 500 ; 0,062.

5. Running thread · Flash [REP] Give the image of $A(2;-3)$ under point reflection in the origin.

Ans. $(-2;3)$.

Sheet 79 Thu. 10 June 2027 Presentation of the specialist subjects; review

1. Warm-up · Flash [NUM] Express $\frac{3}{8}$ as a percentage.

Ans. 37,5 %.

2. Core · Metacognition [MET] What are your three revision priorities before Première?

Ans. Individual answer: find in the year's sheets the questions where you hesitated most.

3. Core + · Flash [EQ] Solve $x^2 = 18$.

Ans. $x^2 = 9 : x = 3$ or $x = -3$.

4. Back · Trap [PUI] Calculate $(-2)^4$ and -2^4.

Ans. 16 and -16.

The brackets decide what the exponent applies to.

5. Running thread · True/false [DIV] True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.

Ans. False. Counterexample: 12 is divisible by 4 and by 6, not by 24; it is divisible by LCM $(4\,;\,6) = 12$.

Annual review and consolidation of Seconde learning

Sheet 80 Mon. 14 June 2027 Annual review and consolidation

1. Warm-up · Flash [GEO] $A(-1;4)$ and $B(3;1)$. Coordinates of $\overrightarrow{AB}$?

Ans. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

2. Core · Method [GEO] Slope-intercept equation of (AB) with $A(0;1)$ and $B(2;5)$.

Ans. Gradient 2; $y = 2x + 1$.

3. Core + · Method [REP] $A(1;0)$, $B(4;1)$, $C(5;4)$. Determine D so that $ABCD$ is a parallelogram.

Ans. $D = A + C - B = (2;3)$; check: $\overrightarrow{AB} = \overrightarrow{DC} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

4. Back · Flash [CAL] Calculate $-5 + 8 \times (-2)$.

Ans. $-5 - 16 = -21$.

Multiplication takes priority.

5. Running thread · Statement→ formula [REG] The area of a square with side x is a function A of x. Give $A(x)$, a table of values for $x = 1; 2; 3$, and say whether A is proportional to x.

Ans. $A(x) = x^2$; 1; 4; 9; not proportional: $A(2) \neq 2A(1)$.

Sheet 81 Tue. 15 June 2027 Annual review and consolidation

1. Warm-up · Flash [NUM] Write 0.00045 in standard form.

Ans. $4{,}5 \times 10^{-4}$.

2. Core · Flash [NUM] Determine $[-2;3[\cap]0;5]$.

Ans. $]0;3[$.

3. Core + · Method [EQ] Solve $|x - 1| < 4$ and give S as an interval.

Ans. $-4 < x - 1 < 4 : S =]-3;5[$.

4. Back · Expand [DEV] Expand and simplify $(2x - 1)^2$.

Ans. $4x^2 - 4x + 1$.

5. Running thread · Flash [GEO] A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.

Ans. $V = \pi \times 3^2 \times 5 = 45\pi$ cm^3.

Sheet 82 Thu. 17 June 2027 Annual review and consolidation

1. Warm-up · Flash [IMG] $f(x) = x^2 - 4x$. Calculate $f(3)$.

Ans. $9 - 12 = -3$.

2. Core · Flash [EQ] Solve $x^2 = 9$ then $|x| = 2$.

Ans. $x = \pm 3$; $x = \pm 2$.

3. Core + · Method [REG] $f(x) = 2x - 3$ and $g(x) = x + 1$. Solve $f(x) < g(x)$.

Ans. $2x - 3 < x + 1 \Leftrightarrow x < 4 : S =]-\infty; 4[$.

4. Back · Flash [QP] A tap delivers 18 L in 4 min. What volume in 10 min?

Ans. $4{,}5 \times 10 = 45$ L.

5. Running thread · Reading code [ALG] What is the final value of `n` ?

```
n = 0
while n*n < 30:
    n = n + 1
```

Ans. $n = 6$ since $5^2 = 25 < 30$ but $6^2 = 36 \geqslant 30$.

Smallest integer whose square exceeds 30.

Sheet 83 Mon. 21 June 2027 Annual review and consolidation

1. Warm-up · Flash [STA] Mean of 5, 7, 9, 11?

Ans. 8.

2. Core · Method [STA] Data set 1; 3; 4; 6; 8; 9; 11: Q_1, median, Q_3.

Ans. $n = 7$: Q_1 index $\lceil 7/4 \rceil = 2$: 3; median (index 4): 6; Q_3 index $\lceil 21/4 \rceil = 6$: 9.

3. Core + · True/false [STA] True or false? Justify. Adding 10 to each value of a data set leaves its standard deviation unchanged.

Ans. True. The deviations from the mean do not change.

4. Back · Reflection [IMG] How many preimages do -1 and 4 have for $f(x) = x^2$?

Ans. -1: none; 4: two (-2 and 2).

5. Running thread · Prove [LIT] Show that for every real number x : $(x-1)(x+1) + 1 = x^2$.

Ans. $(x-1)(x+1) + 1 = x^2 - 1 + 1 = x^2$.

Sheet 84 Tue. 22 June 2027 Annual review and consolidation

1. Warm-up · Flash [DEV] Expand and simplify $(2x - 3)(x + 4)$.

Ans. $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.

2. Core · Flash [EQ] Solve 3 $x - 7 \leqslant 2x + 5$.

Ans. $x \leqslant 12$; $S =]-\infty; 12]$.

3. Core + · Method [EQ] Solve $(x-1)(x+4) = 0$ then $x^2 = 5$.

Ans. $x = 1$ or $x = -4$; $x = \sqrt{5}$ or $x = -\sqrt{5}$.

4. Back · Reflection [PNP] Subscription: 5 € + 3 € per entry. Is the price proportional to the number of entries?

Ans. No: 1 entry $\mapsto$ 8 €, 2 entries $\mapsto$ 11 $\neq$ 16 €.

5. Running thread · Flash [SOL] A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.

Ans. Slant height $\sqrt{3^2 + 4^2} = 5$ cm; $V = \frac{1}{3}\pi \times 9 \times 4 = 12\pi$ cm^3.

Sheet 85 Thu. 24 June 2027 Annual review and consolidation

1. Warm-up · Flash [PRO] What multiplier corresponds to an 8% increase?

Ans. 1,08.

2. Core · Method [PRO] A 25% increase is followed by a 20% decrease. What is the overall rate?

Ans. $1{,}25 \times 0{,}8 = 1$: no change (0%).

3. Core + · Flash [PRO] A price goes from 5,000 to 5,750 Ar. Rate of change?

Ans. $\dfrac{750}{5\,000} = 15\,\%$.

4. Back · Recognise [PER] What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?

Ans. A rectangle; a disc.

5. Running thread · Problem [EQ] Three times a number, minus 4, equals twice that number plus 5. Write an equation then find the number.

Ans. $3x - 4 = 2x + 5$, hence $x = 9$.
Check: $3 \times 9 - 4 = 23 = 2 \times 9 + 5$.

Sheet 86 Mon. 28 June 2027 Annual review and consolidation

1. Warm-up · Flash [MOD] Direction of variation of $x \mapsto -3x + 2$?

Ans. Strictly decreasing (gradient $-3 < 0$).

2. Core · Method [REG] Minimum and maximum of $f(x) = x^2$ on $[-3;1]$?

Ans. Minimum 0 (at 0); maximum 9 (at -3) ; $f(1) = 1 < 9$.

3. Core + · Method [CMP] For $x > 0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.

Ans. $0 < x < x + 1$ and the reciprocal function is decreasing on $]0;+\infty[$: $\dfrac{1}{x} > \dfrac{1}{x+1}$.

4. Back · Construction steps [CON] Describe in three steps how to construct the image A' of A under reflection in a line (d).

Ans. 1. Perpendicular to (d) through A, meeting (d) at H. 2. Using compasses, mark off HA on the other side of (d). 3. A' is the point obtained: H is the midpoint of $[AA']$.

5. Running thread · Method [DEM] A point I satisfies $IA = IB$. Where is it? Which property?

Ans. On the perp. bisector of $[AB]$ (perpendicular bisector property).

Sheet 87 Tue. 29 June 2027 Annual review and consolidation

1. Warm-up · Flash [PRB] $P(A) = 0.28$. Calculate $P(\overline{A})$.

Ans. 0,72.

2. Core · Method [PRB] 100 people including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.

Ans. $P_F(G) = \dfrac{6}{60} = 0{,}1$; $P_G(F) = \dfrac{6}{10} = 0{,}6$.

3. Core + · Method [PRB] $P(A) = 0{,}4$, $P_A(B) = 0{,}5$, $P_{\overline{A}}(B) = 0.2$. Calculate $P(B)$.

Ans. $0{,}4 \times 0{,}5 + 0{,}6 \times 0{,}2 = 0{,}2 + 0{,}12 = 0{,}32$.

4. Back · Problem [MOD] Volume of a cube with edge x : $V(x) = x^3$. Calculate $V(3)$ and find x such that $V(x) = 64$.

Ans. $V(3) = 27$; $x^3 = 64$, $x = 4$.

5. Running thread · Flash [REP] In which quadrant (numbered I to IV anticlockwise) is the point $M(a;b)$ with $a < 0$ and $b > 0$?

Ans. Quadrant II (top left).

Sheet 88 Thu. 1 st July 2027 Annual review and consolidation

1. Warm-up · Flash [GEO] Hypotenuse of a right-angled triangle with shorter sides 5 and 12?

Ans. $\sqrt{25+144}=13$.

2. Core · Flash [REP] Midpoint of $[AB]$ with $A(-3;2)$ and $B(5;-4)$?

Ans. $\left(\frac{-3+5}{2};\frac{2-4}{2}\right)=(1;-1)$.

3. Core + · Write up [DEM] Show that $A(0;0)$, $B(2;1)$, $C(-1;2)$ form a triangle that is right-angled and isosceles at A.

Ans. $AB^2=5$, $AC^2=5$; $BC^2=9+1=10=AB^2+AC^2$: converse of Pythagoras' theorem.

4. Back · Flash [NUM] Give an approximate value of $\frac{1}{3}$ à 10^{-2} accuracy. Is this number a decimal?

Ans. 0.33. No: the decimal expansion of $\frac{1}{3}$ is non-terminating (0.333 . . .).

5. Running thread · Problem [DIV] What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, greater than 1?

Ans. $1+\text{LCM}\ (4\,;6)=13$.